

Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Security Risks & Data Protection Mock Examination

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Total Marks: 150 Marks

Question 1 (Total: 35 marks)

An online retail business stores large amounts of customer transactions and personal information on its digital systems.

(a) State what is meant by unauthorised access. (1)

Define unauthorised access here...

(b) Identify one statement that correctly describes a botnet. Place a cross [X] in the correct box. (1)

- ☐ **A.** A secure fallback storage space for transaction records
- ☐ **B.** A group of compromised computers used for harmful purposes like spreading malware
- ☐ **C.** A software utility that scrambles network data packets using encryption keys
- ☐ **D.** A data validation check that verifies whether an online user is a human

(c) Criminals use automated software scripts to execute brute-force network entry attacks. Describe how unauthorized users leverage these scripts to target platforms with poor security. (2)

Describe the execution mechanism of automated brute-force scripts...

(d) The retail company fears the threat of a malicious data lockdown.

(i) Define the term malware. (1)

Define malware...

(ii) Explain how a specific type of malware known as ransomware operates to extort money from users. (3)

Explain the operational mechanics and extortion process of ransomware...

(e) Data loss can occur through human error or mechanical runtime failures rather than deliberate criminal intent. State three common causes of accidental deletion of files or entire storage drives. (3)

1. 2. 3.

(f) Online customers are frequently targeted by identity theft traps designed to steal their account passwords.

(i) Define the term phishing. (2)

Define phishing...

(ii) Differentiate between standard phishing, spear phishing, and smishing. (3)

Contrast standard phishing, spear phishing, and SMS-based smishing...

(g) Criminals also deploy an alternative site-spoofing fraud technique known as **pharming**.

(i) Describe how criminals use DNS alteration to execute a pharming attack. (2)

Describe DNS poisoning/alteration in pharming...

(ii) Describe how criminals use typosquatting to misdirect internet traffic, providing an example URL. (3)

Explain typosquatting and include an example of a fraudulent URL...

(h) Define the following internet infrastructure terms sourced from the textbook vocabulary: (2)

Webform:

Define Webform...

Domain Name Server:

Define Domain Name Server (DNS)...

(i) Students review inbound emails to avoid clicking on dangerous hyperlinks or malware attachments. State four warning signs that indicate an email message or attachment might be a malicious threat. (4)

List four email security red flags...

(j) Identify two executable file extensions that pose a significant security risk if received as unverified attachments. (2)

Extension 1: Extension 2:

Question 2 (Total: 30 marks)

Network administrators implement distinct software filters, authentication protocols, and protective layouts to secure private data repositories.

(a) Firewalls control the data travelling into and out of an enterprise network boundary. Describe the criteria a firewall examines and how it applies administrative rules to filter incoming traffic. (3)

Describe packet filtering, rule sets, and inspection traits...

(b) Online systems mask user input string characters on physical displays during login routines.

(i) State the purpose of character masking. (1)

State purpose...

(ii) State four formatting rules a student must follow to create a strong, highly secure password. (4)

1. 2. 3. 4.

(c) Web portals deploy specialized interactive tests to block automated data injection routines.

(i) State the full technical meaning of the acronym CAPTCHA. (1)

Full architectural meaning...

(ii) Explain how a standard CAPTCHA test differentiates human interactions from an automatic software program (bot). (2)

Explain validation technique...

(iii) Describe how a reCAPTCHA system functions differently, highlighting its secondary digital contribution benefit. (2)

Describe structural differences and data digitisation benefits...

(d) Anti-malware utilities continuously scan computer files in real-time to combat active code exploits. Compare the primary protective objectives and execution steps of the following three security tools: (6)

Anti-Virus:

Objectives and mechanisms...

Anti-Adware:

Objectives and mechanisms...

Anti-Spyware:

Objectives and mechanisms...

(e) Define how an anti-virus suite utilizes virus definitions and state the action it takes via quarantine procedures. (3)

Explain signature matching and quarantine isolation...

(f) Secure websites rely on specialized hypertexts to keep online payment communications private.

(i) State the full meaning of the acronym HTTPS. (1)

Full meaning...

(ii) Identify two encryption protocols utilized by HTTPS layer frameworks to secure connections. (2)

Protocol 1: Protocol 2:

(iii) Identify the specific visual indicator a web browser displays in the address bar to prove a site is secure. (1)

Visual marker...

(g) State two access levels an administrator can enforce when configuring folder-level file permissions. (2)

1. 2.

Question 3 (Total: 30 marks)

A business implements systematic fallback storage routines to safeguard core database structures against physical disasters or malware attacks.

(a) Define the core purpose of a data backup routine. (1)

Define core purpose...

(b) Complete the following table by inserting the correct technical descriptions for the three backup models. (6)

Backup Strategy Category	Precise Technical Description (Sourced from Textbook Rules)
1. Full Backup	
2. Differential Backup	
3. Incremental Backup	

(c) Differentiate between the storage logic of an incremental backup and a differential backup regarding how they identify and handle modifications since previous backup cycles. (3)

Contrast logic paths, archive bit settings, and recovery speed variations...

(d) A system administrator develops a comprehensive scheduling roadmap to eliminate human error and data collision conflicts. Explain three architectural guidelines or execution practices that constitute a highly resilient backup strategy. (6)

Practice 1: Practice 2: Practice 3:

(e) Explain why an infrastructure planner should avoid utilizing optical media for long-term data archiving strategies. (2)

Explain physical decay and read/write durability limitations...

(f) Financial platforms utilize a variety of secure transaction routers to manage online payments.

(i) Describe how third-party payment processors function to securely transfer funds between users, stating an example platform from the text. (2)

Describe intermediate processing pipelines...

(ii) Identify four specific pieces of data a consumer must physically extract from a standard bank card and input into a webform when completing a digital payment. (4)

1. 2. 3. 4.

(g) State one advanced online transaction validation protocol used by credit card issuers to authenticate online transactions. (1)

State protocol...

(h) Contactless payment systems utilize short-range wireless waves to clear currency balances without inputting physical passwords.

(i) Identify the core wireless connection protocol that powers contactless bank cards. (1)

Identify protocol standard...

(ii) Explain the security design limit used to prevent large-scale financial fraud if a card reader is utilized maliciously by criminals. (2)

Explain transaction currency caps and step-up verification metrics...

(iii) State a simple physical method users can employ to block weak wireless signals from being intercepted by nearby eavesdroppers. (1)

State physical blocking countermeasure...

Question 4: Technical Terms & Threat Matrix (Total: 30 marks)

An IT security consultant is compiling a security report evaluating online data threat vulnerabilities and matching defense configurations. Complete the missing properties within the threat assessment table below by directly sourcing technical definitions and details from the textbook guidelines. (30 marks, 5 marks per matrix row)

Security Event Domain	Core Operational Threat / Performance Mechanism	Sourced Prevention Countermeasure
Ransomware Attack	Distressing messages lock user software systems and demand payments into anonymous accounts.	(a)
Phishing Attempt	(b)	Anti-malware filtering filters and alert inspection of spelling traits.
Pharming Redirect	Website spoofing executed via technical DNS server alteration or typosquatting errors.	(c)
Automated Bot Registrations	(d)	CAPTCHA / reCAPTCHA tests require distorted text inputs.
Spyware Intrusion	Secret monitoring of computer operations via internal keyloggers capturing user keystrokes.	(e)
Wireless Signal Sniffing	(f)	Wrapping physical card structures inside foil layers.

Question 5: Extended Response Essay (Total: 25 marks)

A regional banking corporation is launching a cloud-integrated customer web portal. The system must process credit card transactions, host customer webmail profiles, sync mobile applications wirelessly via short-proximity signals, and secure administrative servers against external penetrations.

The engineering architects are choosing between two security design roadmaps:

- **Option A:** Implementing an aggressive automated protection layer consisting of gateway firewalls with strict port rules, running real-time reCAPTCHA scripts on all webforms, forcing mandatory complex password changes monthly, and running automated nightly incremental backups replicated across multiple geographical server sites.
- **Option B:** Relying on decentralized desktop security configurations where branch staff manually run anti-malware scanners weekly, utilizing full disk backups written to physical media arrays stored locally inside fireproof safes, and implementing symmetric encryption keys for payment paths without online validation firewalls.

Evaluate both options. You should discuss the impacts of each option on data confidentiality (preventing phishing/pharming compromises), resilience against destructive malware (such as ransomware or automated bots), efficiency of system recovery under accidental data destruction, and operational overheads for the bank's technical support staff. (25)

Plan your answer carefully. Marks will be awarded for structured technical progression, accurate usage of ICT vocabulary from the text, balanced comparative arguments, and an effective final conclusion.

Compose your extended response essay analysis here...

End of Question Bank